

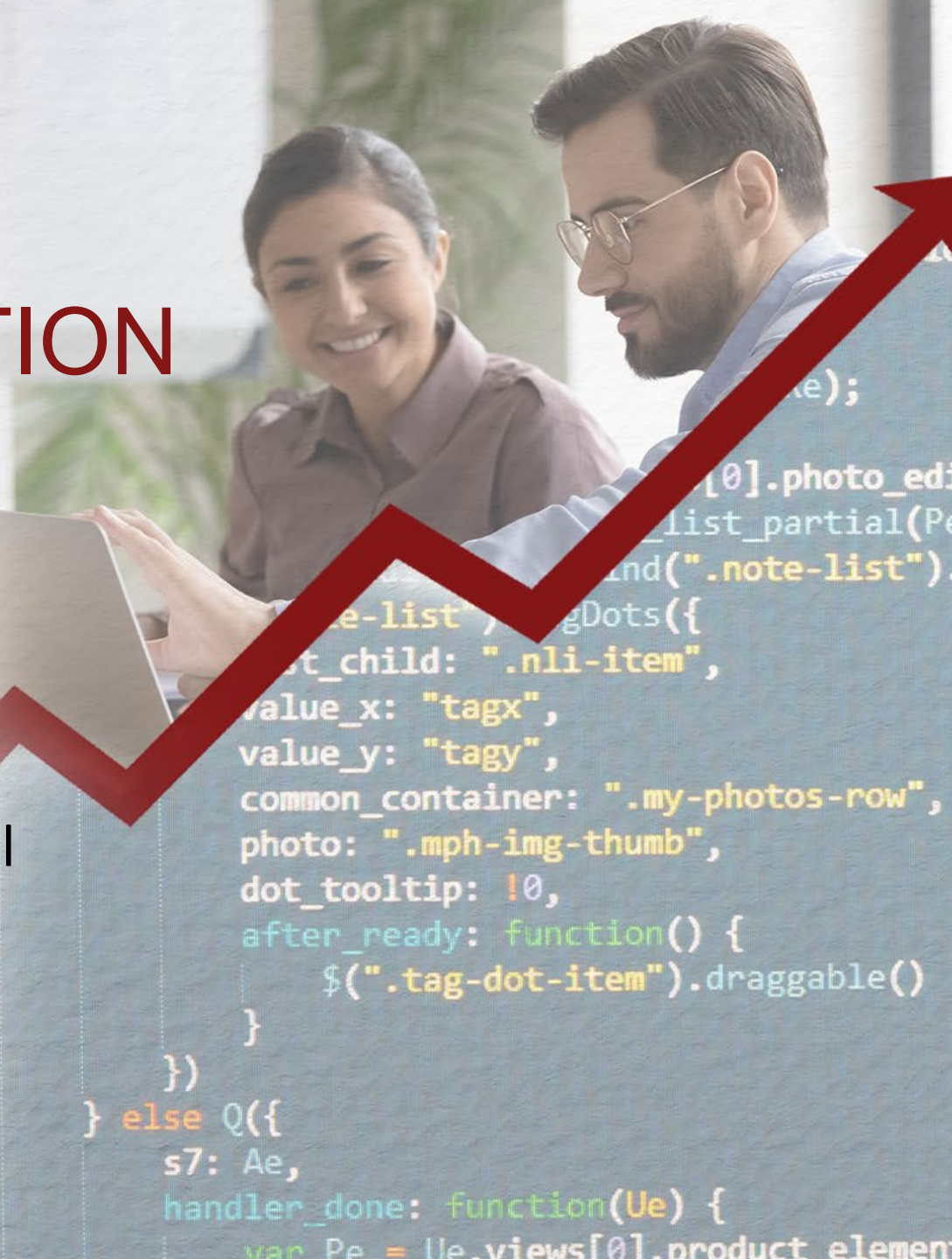
HEALTH DATA VISUALIZATION PROJECT

MIT 5822

Health Information Technologies

Group 1

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Agenda

- Background
- Primary Discoveries
 - Focus on adverse effects by manufacturer
- Implications and Significance of Findings
- Recommendations and Conclusion

Background

- Overall administrative data for comparison was pulled from the Centers for Disease Control (CDC), Kaiser Family Foundation (KFF), and other organizations as indicated on each slide.
- Vaccine Safety data was obtained from the Vaccine Adverse Event Report System (VAERS).
- For this project the data was filtered to complete a retrospective review on 2022 reporting for patient-reported adverse events that were suspected as a result of the COVID-19 Vaccine, first or second dose.

COVID-19 Vaccinations in the United States

Data provided by CDC, final update posted May 11, 2023

[The percent of the population coverage metrics are capped at 95%. Learn how CDC estimates vaccination coverage.](#)

About 81% of the U.S. population received at least one dose of the COVID vaccine. 69% completing the primary series, and about 17% getting an updated Bivalent dose.

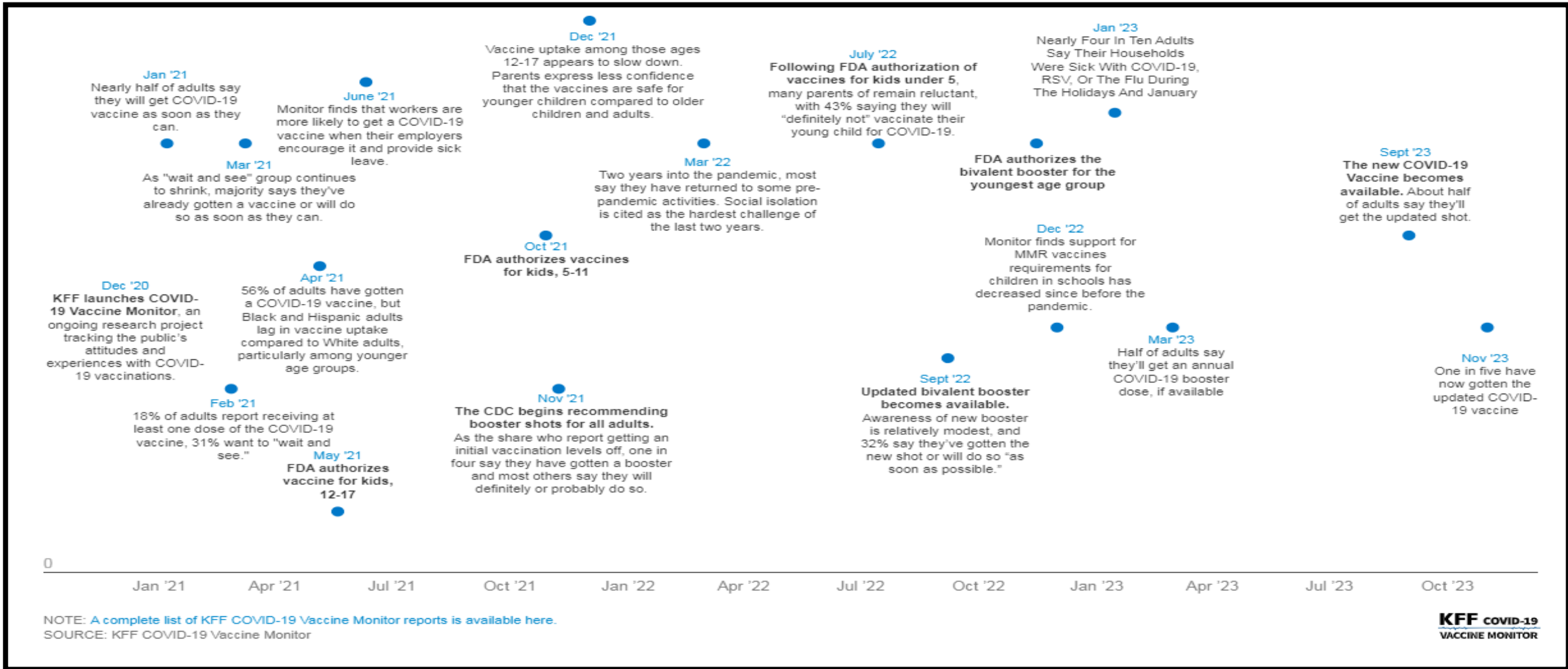
VAERS Data only provides data for people that reported their symptoms related to the vaccine to VAERS.

At Least One Dose	Completed Primary Series	Updated (Bivalent) Booster Dose
Vaccinated People	Count	Percent of U.S. Population
Total	270,227,181	81.4%
Population ≥ 5 Years of Age	268,021,871	85.8%
Population ≥ 12 Years of Age	256,511,884	90.5%
Population ≥ 18 Years of Age	238,239,640	92.3%
Population ≥ 65 Years of Age	58,758,542	95.0%

At Least One Dose	Completed Primary Series	Updated (Bivalent) Booster Dose
People with an Updated (Bivalent) Booster Dose*	Count	Percent of U.S. Population
Total	56,478,510	17.0%
Population ≥ 5 Years of Age	56,352,709	18.0%
Population ≥ 12 Years of Age	54,974,636	19.4%
Population ≥ 18 Years of Age	52,996,306	20.5%
Population ≥ 65 Years of Age	23,699,191	43.3%

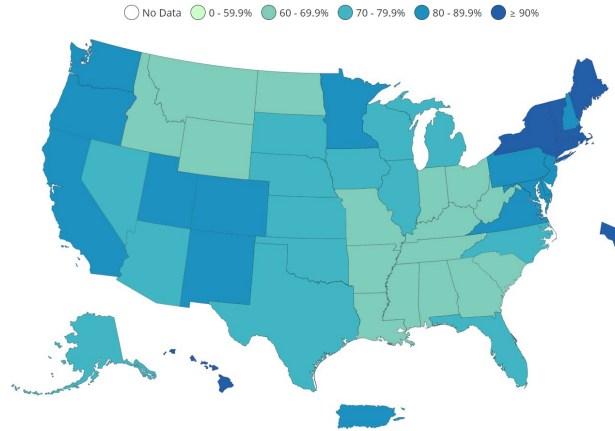
At Least One Dose	Completed Primary Series	Updated (Bivalent) Booster Dose
People Who Completed a Primary Series*	Count	Percent of U.S. Population
Total	230,637,348	69.5%
Population ≥ 5 Years of Age	229,426,936	73.5%
Population ≥ 12 Years of Age	219,966,681	77.6%
Population ≥ 18 Years of Age	204,327,579	79.1%
Population ≥ 65 Years of Age	51,708,613	94.4%

Kaiser Family Foundation Overview

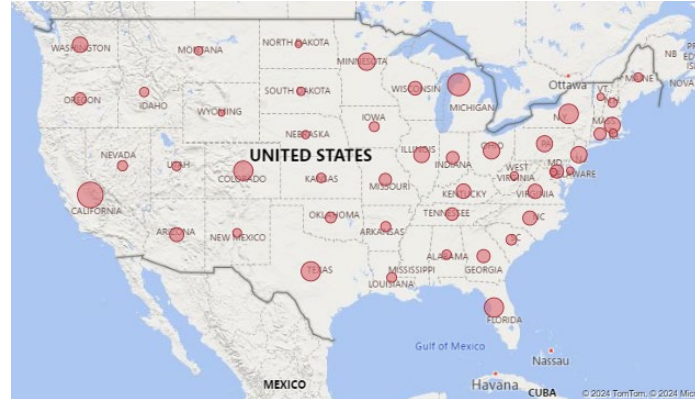


Geographic Relationships

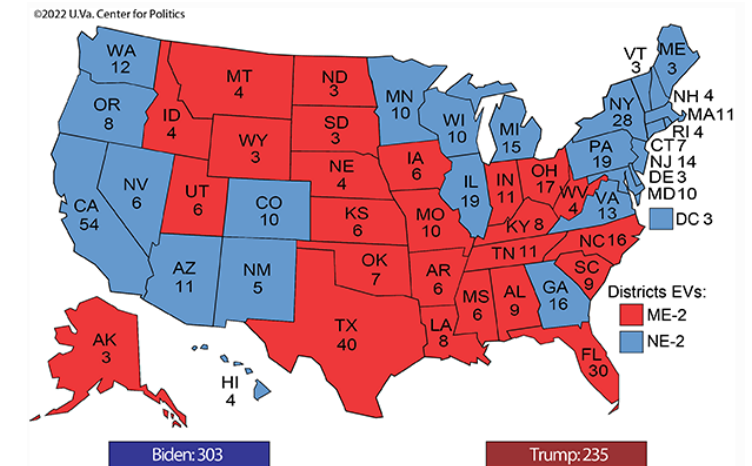
Percent of the Population 18 Years of Age and Older with a Completed Primary Series Reported to CDC by Jurisdictions and Select Federal Entities



The proportion of the Population 18 Years of Age and Older who Reported an Adverse Event through VAERS

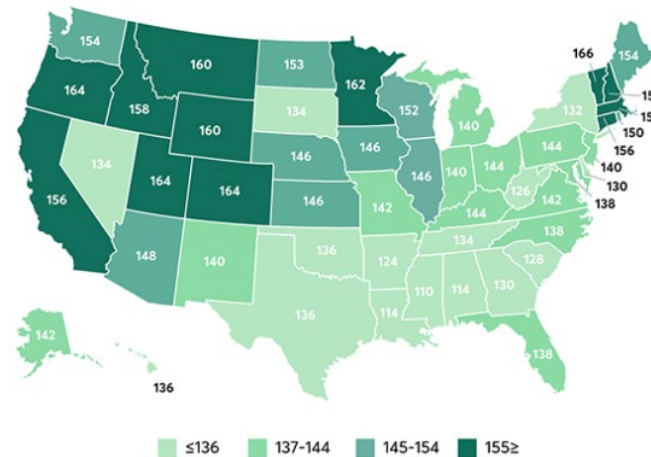


2020 Election Results: Examining Political Connections with Adverse Event Reporting



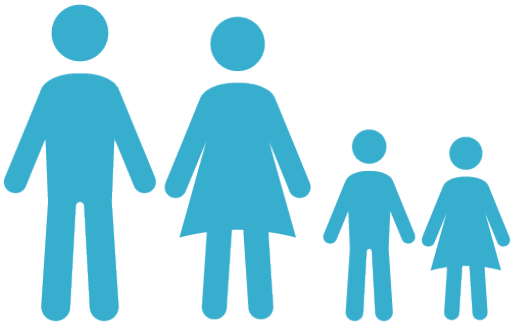
Health Literacy In The 50 States – Key Findings

Median Health IQ by State



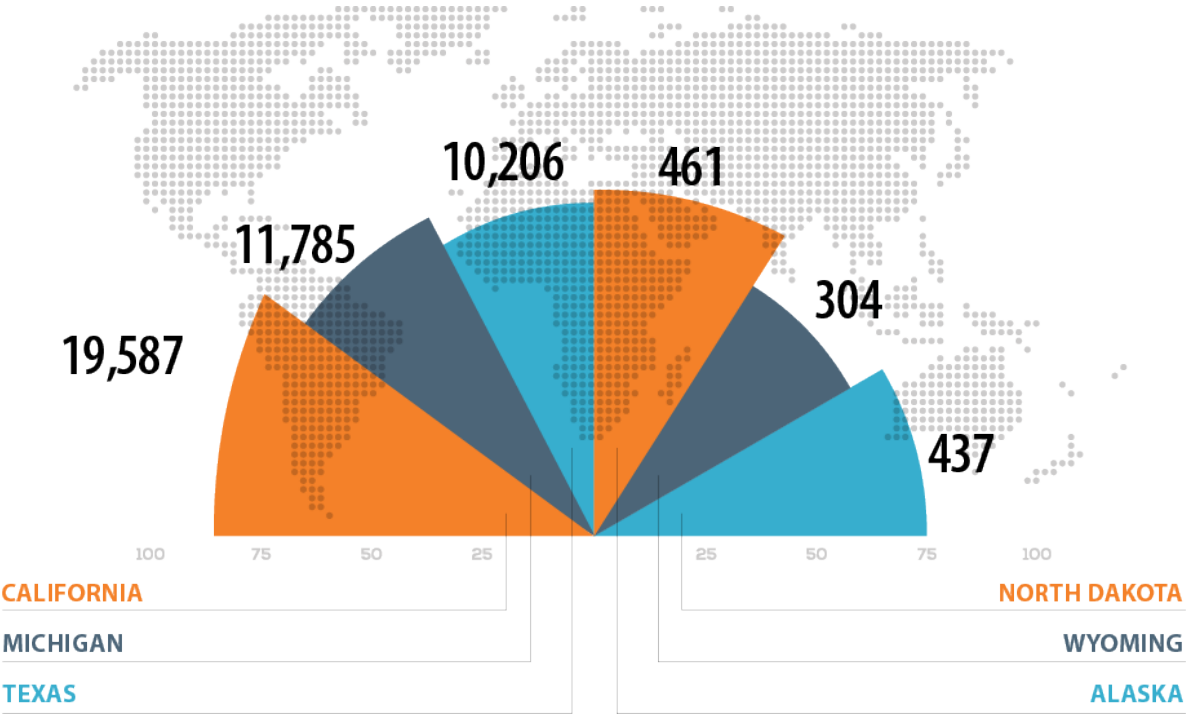
COVID-19 Vaccinations with Adverse Events Reported in VAERS per State

Top Three States with Highest and Lowest Number of COVID-19 Vaccinations with Adverse Events

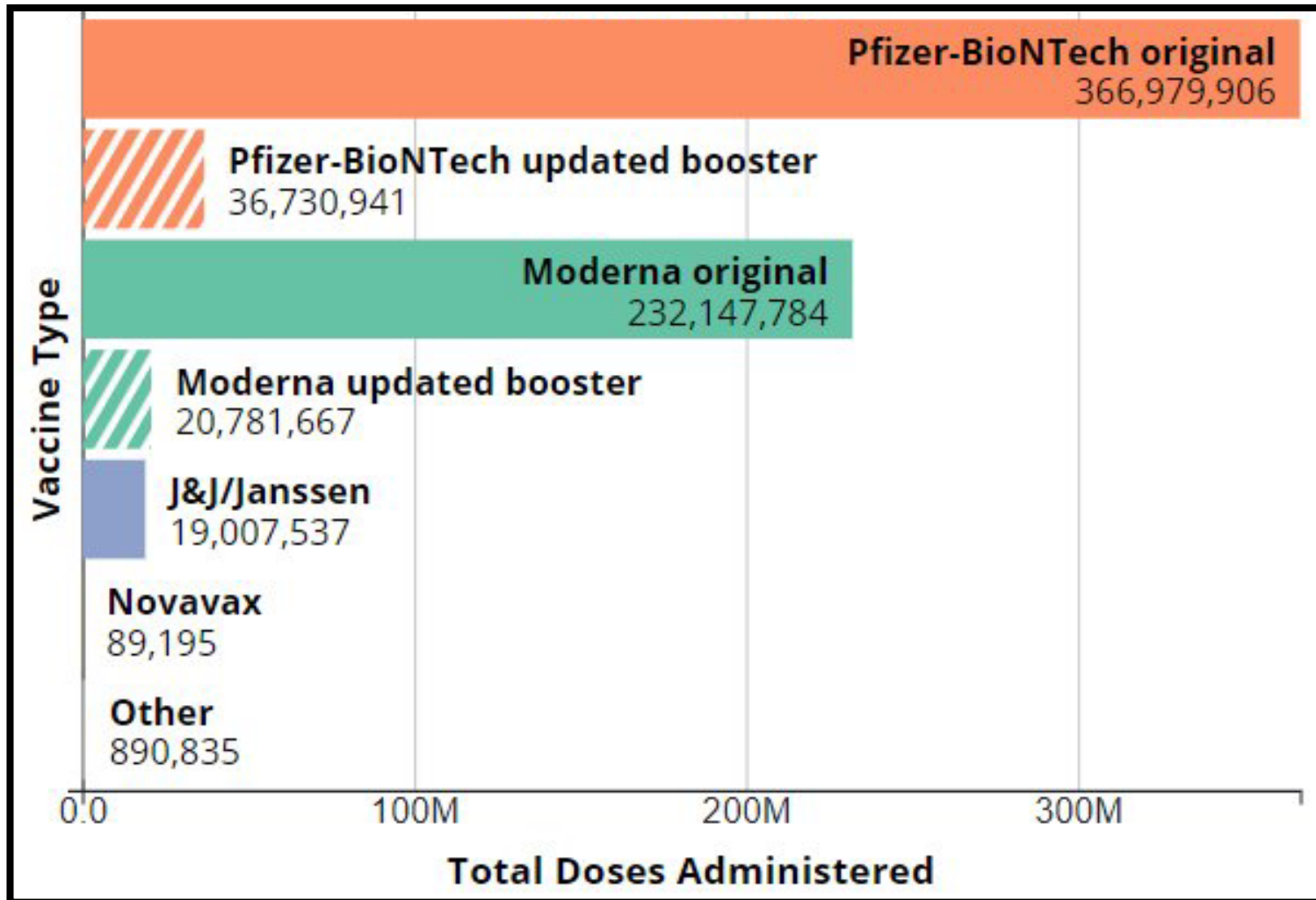


86%

There is death in all age groups. The most affected age group is senior citizens. 86% of deaths are 60 to 90 years old, out of which 53% are males.



U.S. Covid-19 Vaccine Administration by Vaccine Type



Manufacturer Specific Trends

The data shows Pfizer/BioNTech leading in adverse event reports, followed by Moderna, Janssen, unknown manufacturers, and Novavax. This distribution can inform several layers of analysis:

1. Vaccination Volume Correlation:

- The number of reports of adverse events is likely linked to the total number of doses given for each vaccine. Since Pfizer/BioNTech and Moderna are the first vaccines available and have been widely used in many regions, their report counts are naturally higher. It is essential to consider this aspect while looking at the data. Higher report numbers do not necessarily mean a higher risk of adverse events but rather a larger population that has been vaccinated.

2. Public Reporting Bias:

- Variations in reporting rates between vaccines may reflect public awareness and reporting behavior. High-profile news coverage or social media discussions about specific vaccines could influence individuals to report adverse events.

3. Safety Profile and Vaccine Technology:

- Adverse event reports serve as a valuable data source for continuous safety monitoring. Various vaccine technologies such as mRNA vaccines like Pfizer/BioNTech and Moderna and viral vector vaccines like Janssen may have unique safety profiles, which could be reflected in the types and frequency of adverse events reported.
- Analysis of adverse events by manufacturer can identify safety profiles and specific vaccine reactions.

Population Distribution of Vaccine Adverse Events by Vaccine Schedule

211.07K

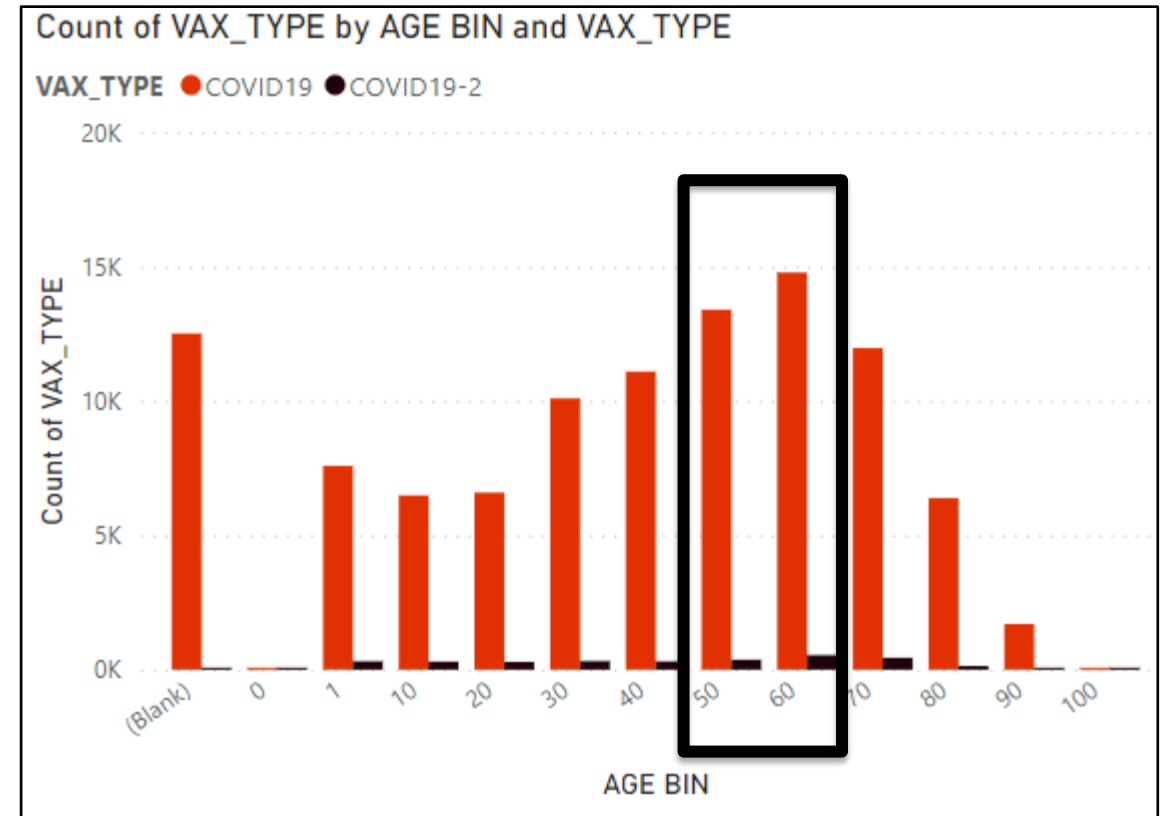
TOTAL INDIVIDUALS W/ ADVERSE EVENTS

49.31

Average of AGE_YRS

Key Findings

- Most reported adverse events occurred with the first dose of the Covid-19 vaccine despite 85% of Americans completing the Primary Covid Vaccine Series.
- 50-60s were impacted most by the first dose and second dose.



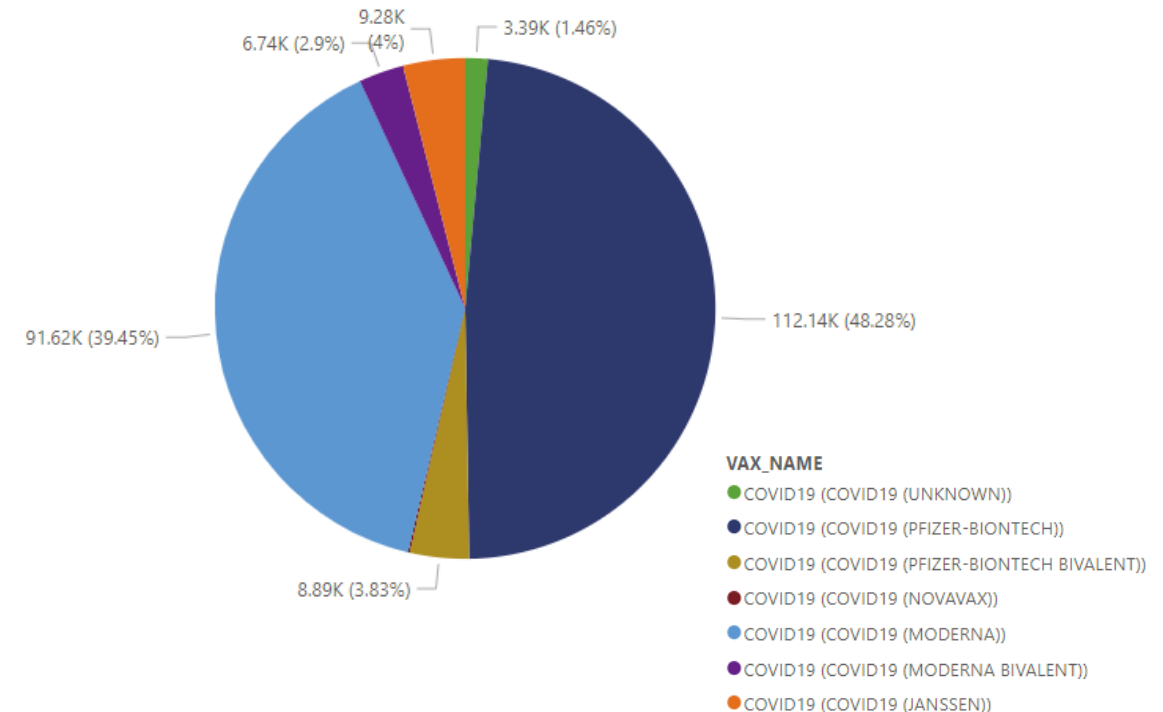
Severity of Events by Manufacturer

Key Findings

- The severity rates must be contextualized against the backdrop of total vaccination numbers to accurately assess risk.
- It should also be noted that there is a chance for underreporting due to lack of public awareness around VAERS or provider unwillingness to submit VAERS reports for suspected adverse event due to legal or malpractice cautions, or personal beliefs.
- 52% of reported adverse events were Pfizer-Biontech followed by Moderna at 42%. Overall administration of Pfizer was around 60%, and Moderna at 37% of total vaccines administered.
- There are differences in the proportion of severe adverse events (hospitalizations, life-threatening conditions, deaths) among vaccines from Pfizer/BioNTech, Moderna, and Janssen.

VAX_NAME	Count of VAERS_ID
COVID19 (COVID19 (JANSSEN))	9280
COVID19 (COVID19 (MODERNA BIVALENT))	6744
COVID19 (COVID19 (MODERNA))	91620
COVID19 (COVID19 (NOVAVAX))	210
COVID19 (COVID19 (PFIZER-BIONTECH BIVALENT))	8889
COVID19 (COVID19 (PFIZER-BIONTECH))	112139
COVID19 (COVID19 (UNKNOWN))	3388
Total	232270

Count of VAERS_ID by VAX_NAME



Types of Adverse Events by Manufacturer

Background

- Different COVID-19 vaccines have been developed by various manufacturers using various methods, which may result in distinct adverse event patterns reported after vaccination.
- The next slides provides a detailed analysis of the adverse events associated with three leading COVID-19 vaccines: Pfizer/BioNTech, Moderna, and Janssen, as indicated by the CDC.
- By comparing the most common adverse events reported for each vaccine, we aim to provide healthcare professionals with insights to help manage patient expectations and address their concerns more effectively.

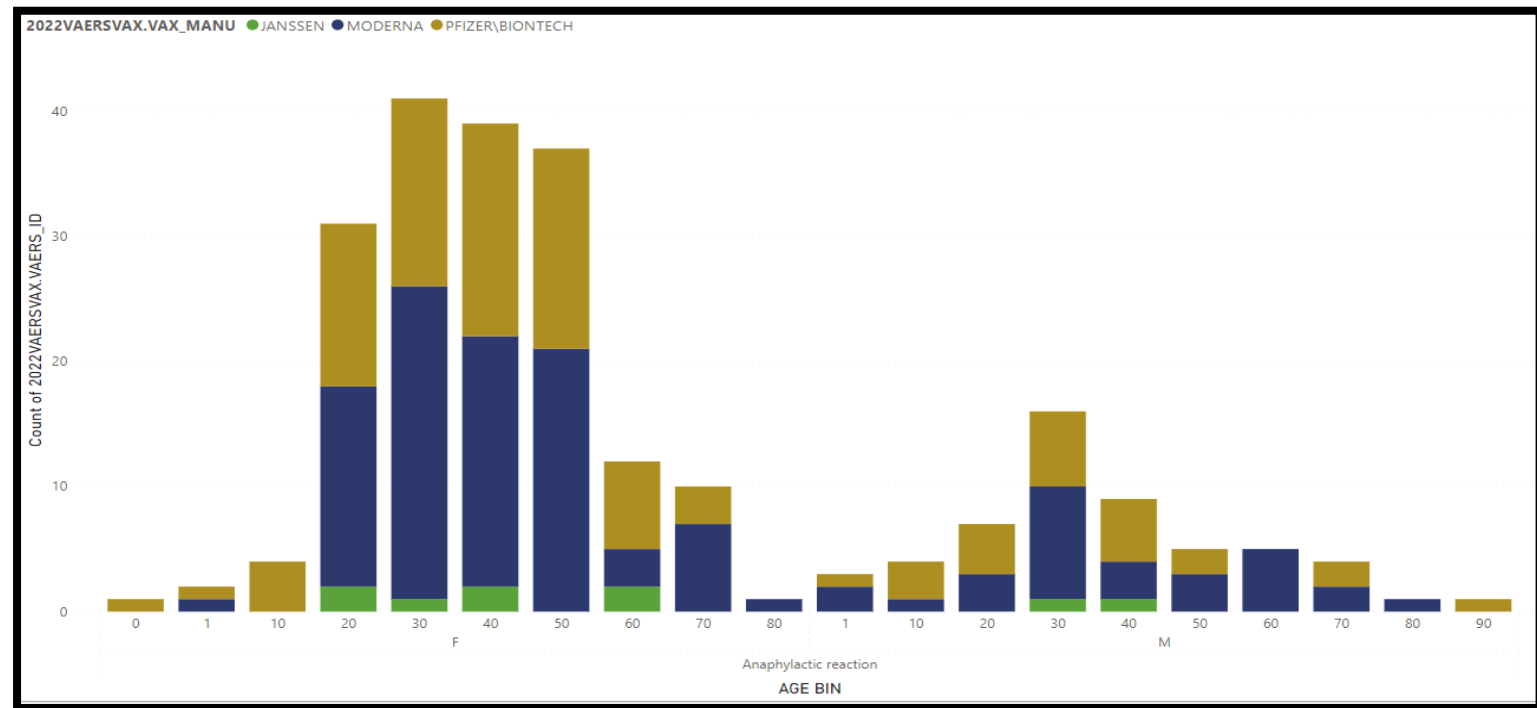
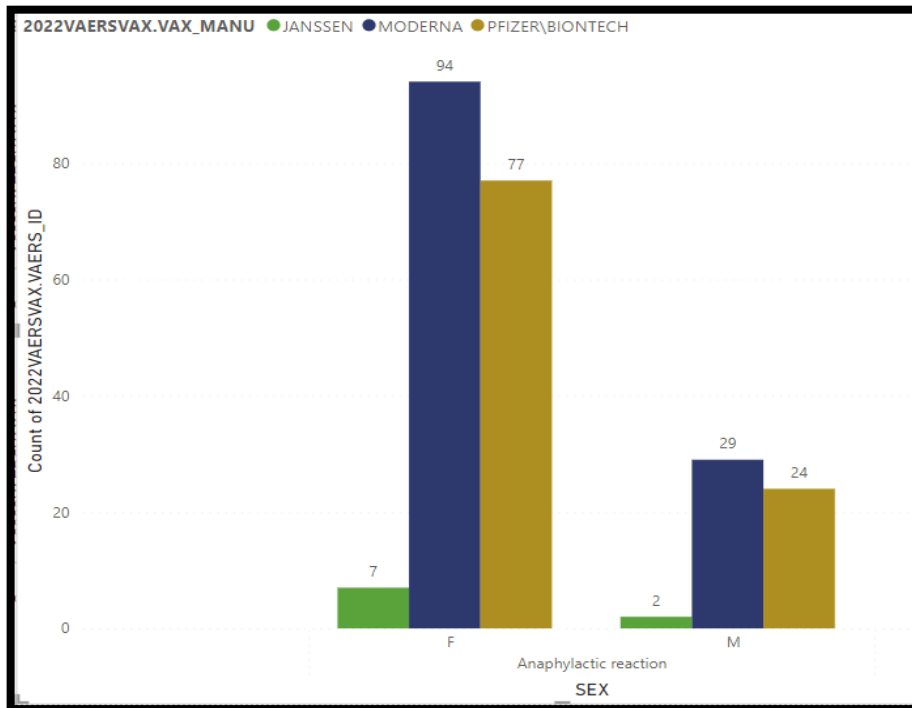
Severe Adverse Events Analyzed

- Anaphylactic Reaction
- Guillain-Barré
- Myocarditis

Anaphylactic Reaction

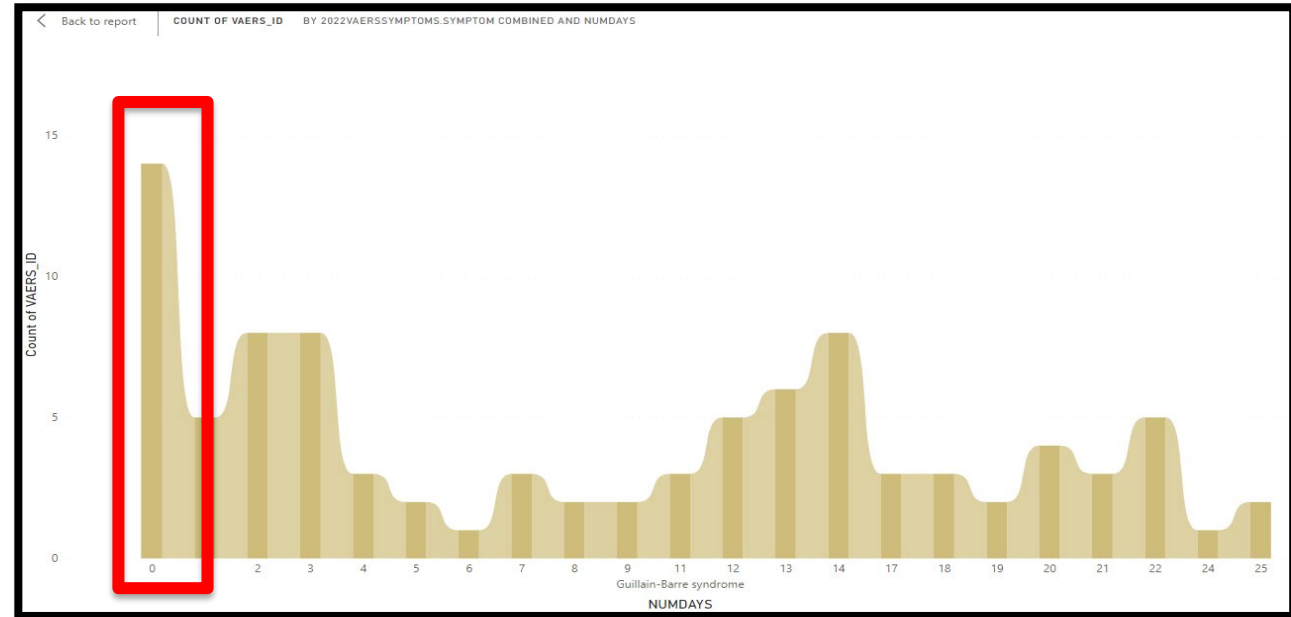
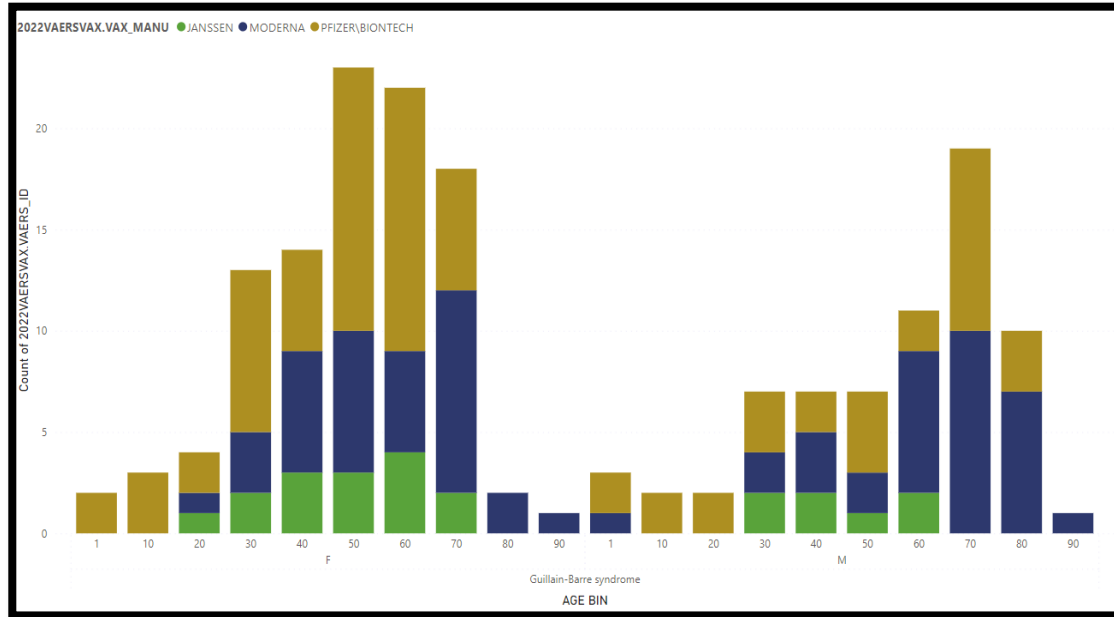
Severe, life-threatening allergic reaction

- According to CDC, occurred at a rate of approximately 5 cases per one million vaccine doses administered - 0.0005%.
- VAERS data reported 234 occurrences, about .0002% of doses administered. Indicates underreporting.
 - Females were affected 76%, and most effected in their 30s and 40s
 - Men were affected 23% and most effected in 30s and 40s .
 - Moderna accounted for 52% of occurrences, then Pfizer/BioNtech at 43%, and Janssen at 3%.



Guillain-Barré Syndrome (GBS)

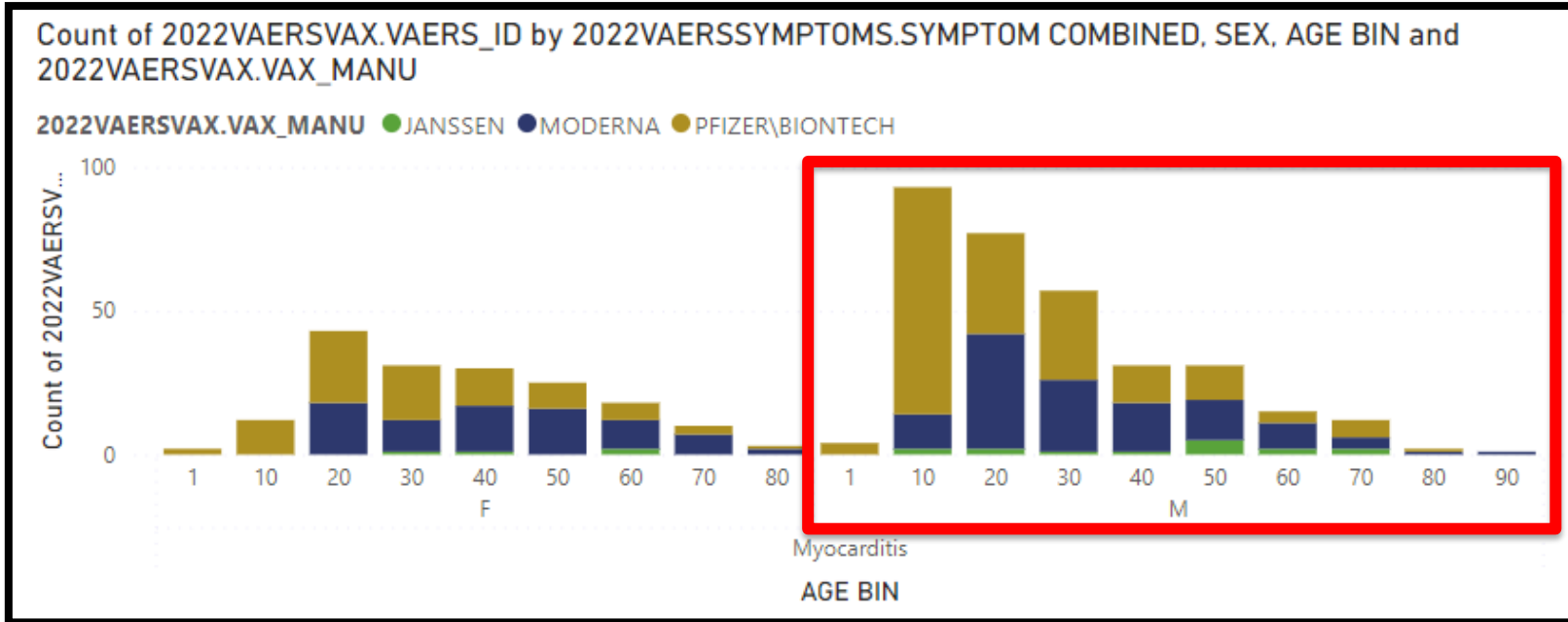
Body's immune system damages nerve cells, causing muscle weakness and sometimes paralysis.



- Females were affected 76% vs 24% for males, highest affected in ages 50 years or older.
- Vaccine Safety Datalink (VSD) indicated increased rate of GBS within 21 days after Janssen vaccine, with no increased risk indicated from Pfizer/BioNtech or Moderna.
- Analyzed VAERS data did not show that, but it did indicate onset was reported to most likely occur on the date any COVID19 Vaccination was received.

Myocarditis

Inflammation of the heart muscle



Most significant prevalence was males ages 10-19. Why is this?

Possible factors include:

- Genetic predisposition/hormonal factors
- Underlying conditions
- Reporting factors

There were no significant findings from VAERS to indicate correlation with geolocation, allergies, # of hospital day stays.

- Males were reported with myocarditis significantly more – 65% males vs 35% females.
 - Males ages 10-30 were most affected. Women ages 20-40 most affected.

Temporal Trends in Adverse Event Reporting

Background

This analysis examines the temporal patterns of adverse event reports related to COVID-19 vaccines, revealing how these reports have fluctuated during the vaccine rollout.

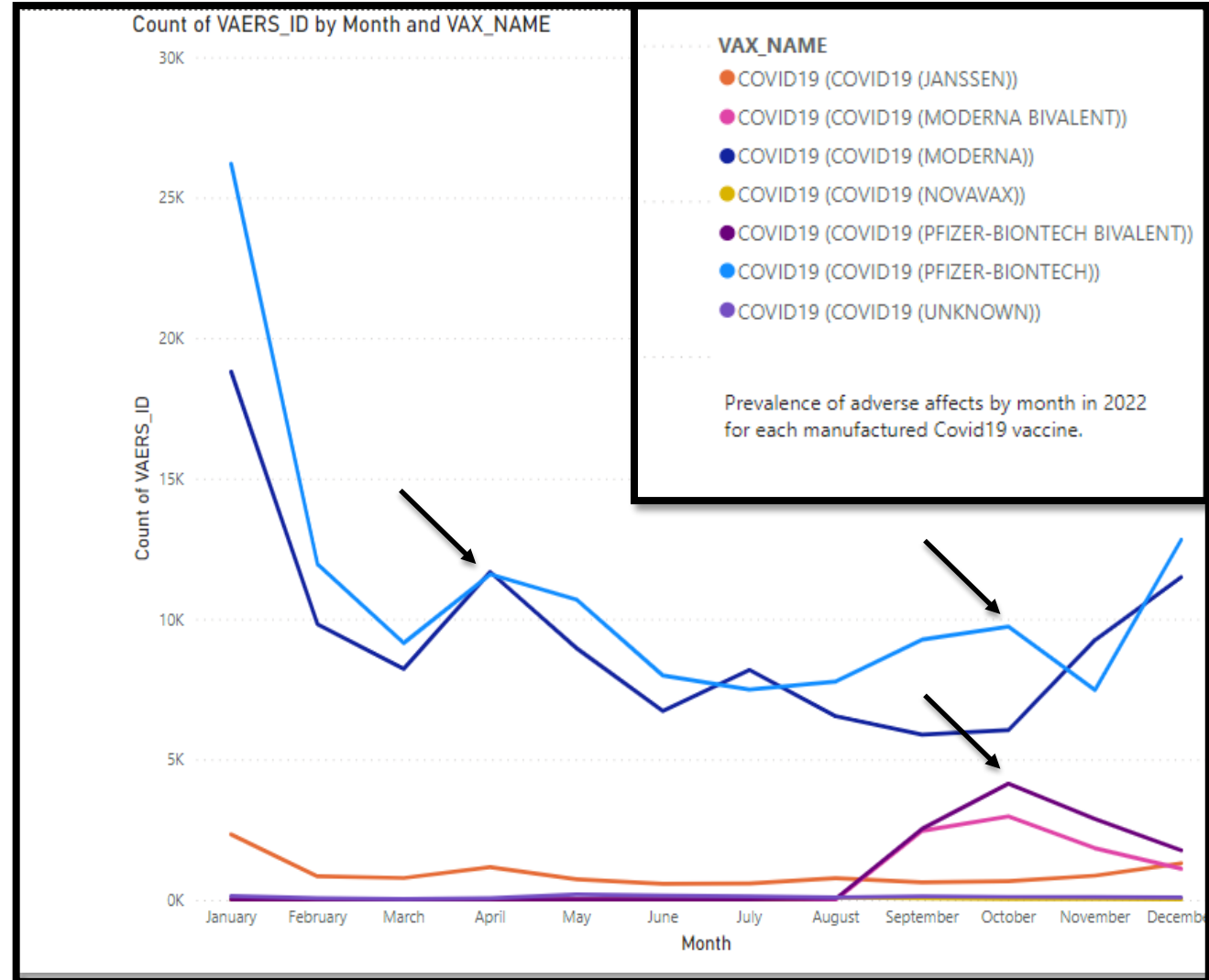
Key Findings

April Peak

- **Vaccine Rollouts:** This peak could coincide with a new phase in the vaccine rollout, such as eligibility expansion or the introduction of booster shots.
- **Public Awareness:** Could have seen increased public awareness due to media coverage or public health campaigns, potentially increasing the rate of adverse event reporting.
- **Regulatory Decisions:** Any safety communications or regulatory decisions made in response to safety data might have resulted in a heightened vigilance for reporting adverse events.

October Peak

- **Booster Campaigns:** There might have been booster campaigns in response to waning immunity, which could lead to a fresh wave of vaccinations and subsequent reports.
- **Seasonal Factors:** Autumn months often see an uptick in vaccination efforts in preparation for winter season, where respiratory illnesses including COVID-19 tend to rise.
- **Awareness and Reporting:** May also be attributed to increased reporting following heightened public discourse about vaccine safety, possibly influenced by the previous peak's investigations and findings.



Why These Trends Matter: Implications

1. For Public Health Officials:

- The trends in adverse event reporting provide a real-time indication of the safety of vaccines. This information helps in making policy adjustments and targeted interventions. By identifying patterns in adverse event reporting, potential safety signals can be quickly identified and managed. This ensures that the benefits of vaccination programs continue to outweigh the risks.

2. For Healthcare Providers:

- By understanding common adverse events and their temporal fluctuations, healthcare professionals can communicate better with patients, creating transparency, trust, and empowerment when making informed decisions about their health.

3. For Vaccine Manufacturers:

- It is critical to understand adverse event reporting to improve vaccine safety profiles, refine vaccine formulations, and inform future clinical trials. For example, different vaccines require different temperature ranges – maintenance of supply chain for this variation is crucial.

4. For the General Public:

- Clear communication on vaccine safety combats misinformation and reinforces participation in vaccination efforts. Educating the public on systems such as VAERS improves data efficacy and trust for all.

Recommendations: Addressing Health Equity and Bias

Why Do We Need Health Population Data?

- **To identify social determinants of Health:** Data collection across diverse groups helps identify health disparities and the underlying factors contributing to them.
- **To understand health disparities:** Different populations may experience variations in health outcomes that require a specified treatment plan.
- **To measure health system performance:** Assessing the performance of healthcare systems requires data that reflects the diversity of the population served. This information is essential for evaluating the effectiveness of healthcare services and interventions across different communities.

Capturing Health Data Improves:

- **Public Health Planning and Policy:** Capturing accurate health data from various populations helps authorities allocate resources, implement targeted interventions, and create policies specific to diverse communities.
- **Research and Scientific Advancements:** Including diverse populations in health research facilitates a more comprehensive understanding of diseases and their treatment. It enables researchers to identify genetic, environmental, and social determinants of health that may vary among different ethnic groups.
- **Identifying Social Determinants of Health:** Health outcomes are influenced by social determinants such as socioeconomic status, education, and living conditions, which can vary across populations. Collecting data on different ethnic groups helps identify and address these social determinants to improve overall health.



Recommendations: Data Quality and Use

Strategies for Enhancing the Quality of Health Data

- **More discreet and structured data fields:** Free-text entries introduce ambiguity and limit data quality. Data should be consistently captured and comprehensible.
- **Data Literacy with Layperson-Friendly Examples:** To enhance effective contribution to valuable health data and enrich databases with relevant data.
- **Mandate Provider Reporting for Adverse Events:** Distinguish “suspected” and “confirmed” flags to aid in risk assessment, foster transparency in data reporting, and ensure comprehensive data collection.
- **Provide Close-Loop Resources and Referrals System:** For patients and providers to report adverse events in VAERS that link to vaccine manufacturers, education and peer support, and the National Vaccine Injury Compensation Program (VICP)
- **Public Education on VAERS:** By launching educational campaigns to inform the general public about the VAERS reporting system to improve the effectiveness of public health initiatives.
- **Use LLM to Produce Better Public-Facing Reporting:**



Questions?

